



HUNGARY

ROBOSYSTeAM



Mentor



Mentor



Team Leader
Programming
Hardware



Hardware
Design



Design
Documentation



Purchasing
Documentation

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SCAN ME

Introduction:

We, the members of the **RoboSysTeam**, prepared for this year's competition with great enthusiasm. This is already the second year that we are participating in the RoboCup Junior Soccer Lightweight category.

The composition of our team remained the same; however, during the past year we gained a great deal of new experience and knowledge in the fields of robotics, programming, and engineering thinking. With the support of our mentors, Tibor Pacsuta and László Jurás, we continuously worked on improving our robots.

Our goal was to build a reliable, fast, and efficient robot that is capable of reacting autonomously to events occurring on the field. For us, the preparation was not only a technical challenge, but also a shared learning process. During our teamwork, many new ideas were born, our problem-solving skills improved, and we gained a great deal of valuable experience that we will also be able to use in the future.

Contact Info:

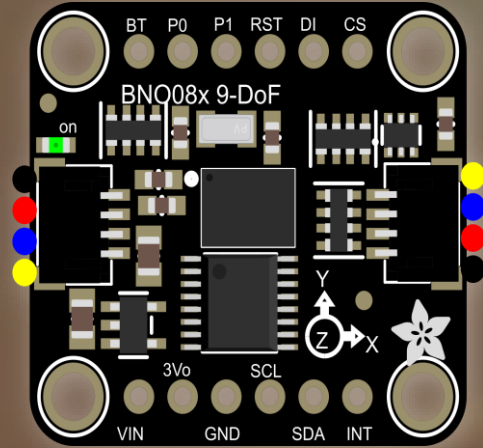
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Compass

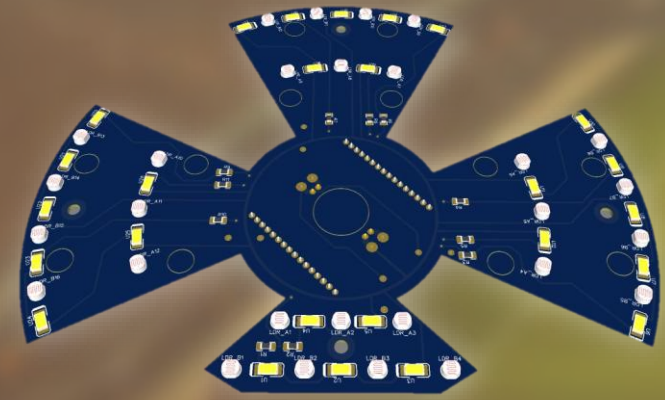


The robot's orientation is ensured by a **BNO085 IMU** sensor, which provides **real-time position and rotation data**. With the help of this sensor, the system continuously monitors the robot's current heading and compares it to the desired direction of movement. The built-in control algorithm automatically corrects any deviations, allowing the robot to maintain a stable orientation while moving, even in the event of collisions or sudden changes in direction. This is especially important for **accurate ball handling and aiming**, as it ensures that the robot is always facing the correct direction during gameplay.

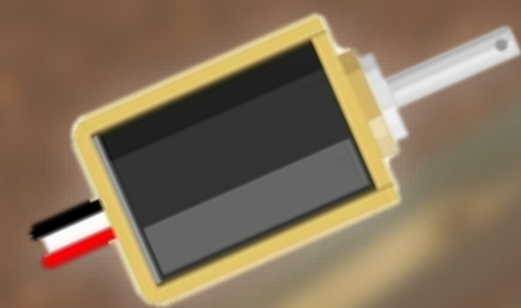
Line Avoiding:



The robot's line detection system is based on a custom printed circuit board designed using the EasyEDA software. The board includes **28 photoresistors**, 26 LEDs, and two 16-channel multiplexers. The multiplexers transmit the sensor data to the **Teensy 4.1 microcontroller**, where the software organizes and analyzes the signals in groups. As a result, the robot is capable of accurately determining its position near the edges of the field or along the white boundary lines, while intelligent motor control ensures that it reliably remains within the playing area.



Kicker Mechanism and Ultrasonic Sensing:

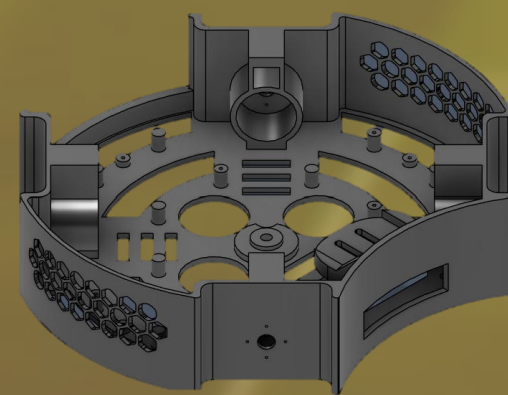


The robot's ball-handling is ensured by a fast-response, electrically actuated kicker mechanism capable of propelling the ball at high speed in a short time. The operation of the system is supported by a downward-facing ultrasonic distance sensor, which continuously measures the position of the ball within the receiving area. Using the ultrasonic sensor, the robot precisely determines whether the ball is in the correct position for kicking. If the measured distance falls outside a predefined range, the control unit activates the kicker. This solution minimizes incorrect or premature kicks.

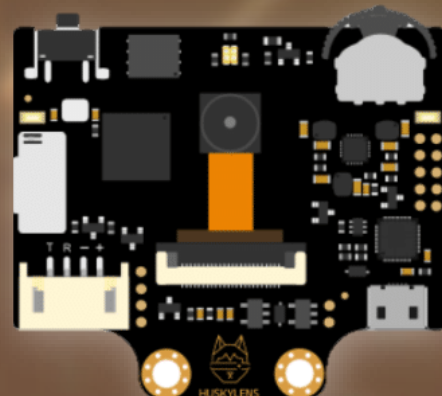
Frame Structure Design:



When designing the frame structure of our robot, our goal was to create a compact, lightweight, yet stable construction. The design was carried out using **Autodesk Fusion 360 CAD software**, which allowed us to test and optimize several different configurations. Compared to last year's robot, we developed a smaller, multi-level frame that makes more efficient use of the available space. This structure ensures that all sensors and electronic components are arranged in an organized and well-positioned manner within the robot.

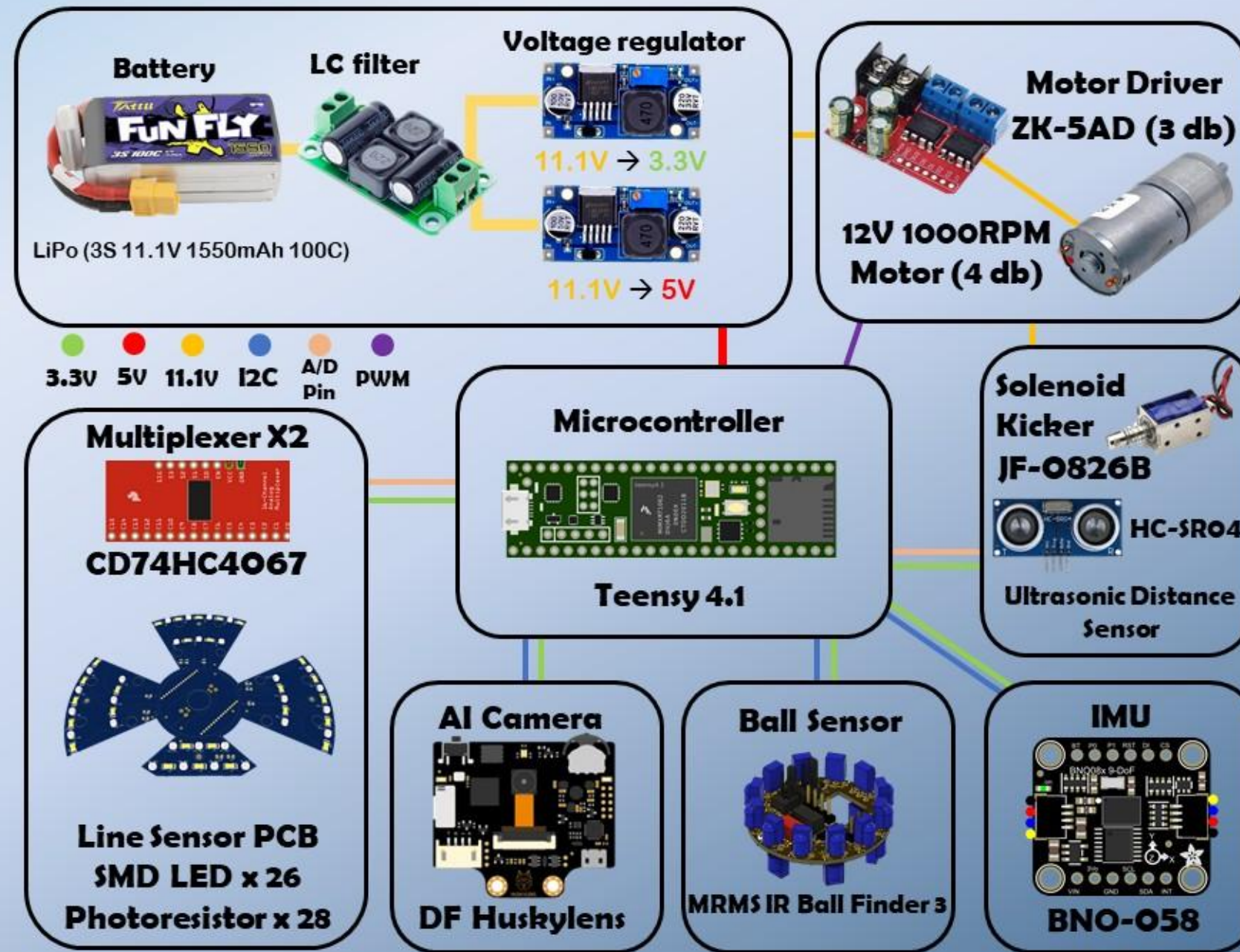


AI-based goal detection using a camera:



The robot implements goal detection and positioning using a **HuskyLens AI Camera**. The built-in artificial intelligence algorithms enable reliable real-time identification of the goal, even under varying lighting conditions and fast-paced game situations. Based on the positional data provided by the camera, the robot continuously adjusts its orientation and turns toward the more open areas of the goal. This increases shooting efficiency, as the system also takes into account the optimal aiming point of the goal.

Robot Structure

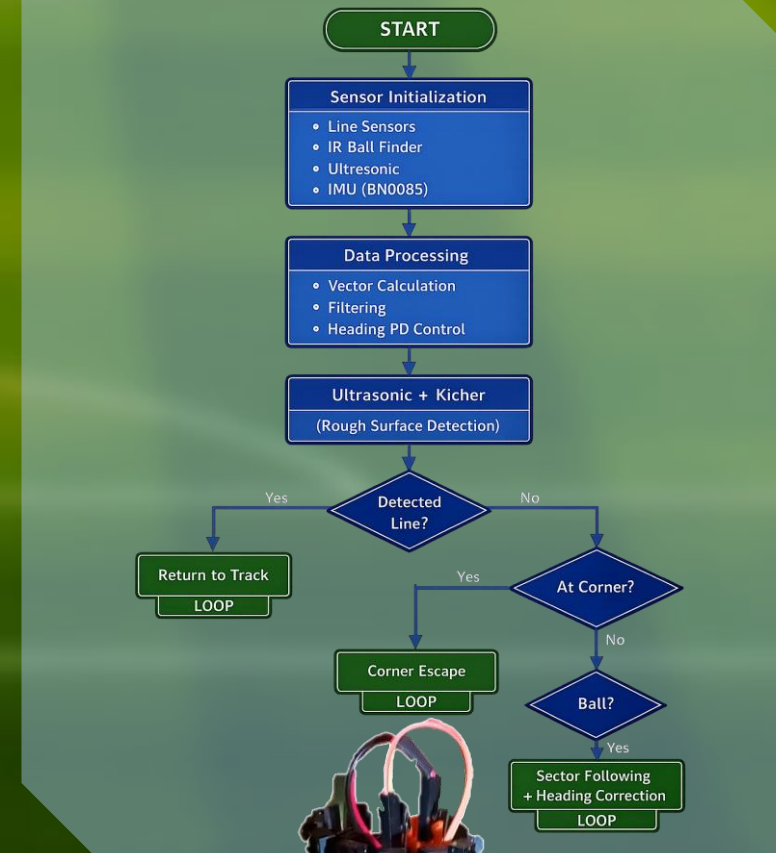


CONTROL SYSTEM

The robot operates based on a modular, sensor-based decision-making system that processes data from multiple sensors in real time. Central control is provided by a high-performance Teensy 4.1 microcontroller, which handles motion, sensors, and decision-making in parallel. The system operates in a cyclic manner: in each loop, it reads the line sensors, the IR Ball Finder module, the ultrasonic sensor, and the BNO085 IMU. During data processing, it determines the direction and distance of the ball, while a PD controller stabilizes heading control.

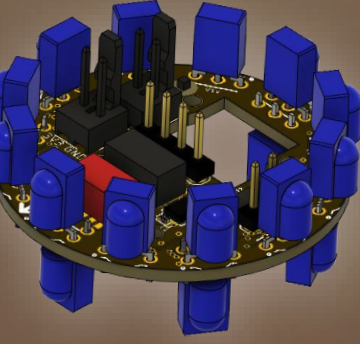
The decision-making process is hierarchical: first maintaining position on the field, then handling corner situations, and finally ball tracking. If the ball reaches the correct position, the ultrasonic sensor triggers the kicker mechanism. This structure ensures fast and stable operation in changing environments. In addition, the system applies noise filtering and error handling to minimize issues caused by sensor uncertainty.

The modular architecture allows for **easy integration** of new sensors or functions in the future. As a result, the robot is easily upgradable and adaptable to different competitive environments.

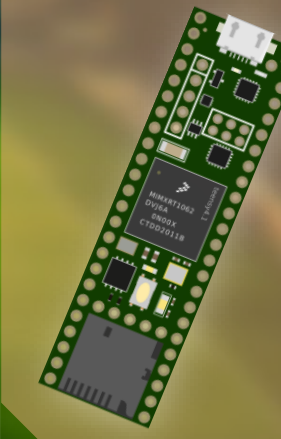


Ball Detection:

To determine the position of the ball, we use the **MRMS IR Ball Finder 3** sensor. The device is capable of determining the direction of the ball relative to the robot by detecting infrared signals. The sensor data is read via I2C communication, which enables fast and stable data transfer, allowing the system to react to changes in game situations in real time. The incoming signals are interpreted using weighted processing and filtering, which reduces noise and increases measurement accuracy. The resulting angle is divided into different directional ranges, each associated with predefined motion patterns. As a result, the robot not only detects the position of the ball but is also able to move toward it dynamically and stably, ensuring continuous and efficient ball tracking on the field.



Software:



The program is developed using the Arduino IDE development environment, which allows for rapid testing and continuous improvement of the robot's software. To handle the robot's sensors, we use several different libraries and communication methods. For compass and orientation measurement, we use the Adafruit BNO08x Library, which helps maintain stable directional control of the robot on the field. Communication with individual sensors and peripherals is implemented via the I2C protocol, ensuring fast and reliable data transfer between the robot's various systems. During development, significant emphasis was placed on optimizing algorithms and filtering sensor data to ensure that the robot operates as stably and accurately as possible during matches.

Robot Motion:



The robot is driven by **four 1000 rpm, 12V motors**, which provide the necessary force for movement. These are controlled through two ZK-5AD dual H-bridge motor drivers. The GTF Omni Wheels enable fast, precise, and unobstructed directional changes. The speed and rotation direction of the motors are regulated by the **PWM outputs** of the Teensy 4.1 microcontroller, ensuring stable and accurate movement on the field.

Power Supply:

The robot's power supply is provided by a **11.1 V, 1550 mAh lithium polymer battery**. For stable operation, we use two LM2596 DC-DC buck converters, which provide regulated 5V and 3.3V outputs for the sensors and control systems.



Future developments:

During further development of the robot, we plan to **integrate a dribbler mechanism**, which will enable more stable ball handling and more precise control. In addition, our goal is to implement faster data processing, which reduces reaction time and improves the efficiency of decision-making. To increase system performance, we also plan to **use stronger and faster motors**, which will provide more dynamic and efficient movement.

Budget (1 Robot)

Motor (4 db)	140€
Omni Wheel (4 db)	170€
Motor driver (3 db)	10€
Teensy 4.1	60€
BNO085	38€
MRMS Ir Ball Finder 3	38€
Photoresistor (28 db) + SMD Led (26 db)	25€
Huskylens AI camera	75€
Solenoid kicker + Ultrasonic distance sensor	18€
PCB panel	60€
Battery	28€
PLA Filament	23€
Other costs	125€
Total	810 €